

Dataset format: .csv

To balance the classes in UNSW-NB15, the proposed CTGAN generated a total of 215,000 synthetic samples for ten classes. The synthetic samples are divided into 40,000 ‘Normal’ class samples; 35,000 samples each for both ‘Worms’ and ‘Analysis’ classes; 25,000 samples for ‘Backdoor’; 20,000 for ‘Exploits’; 15,000 samples each for ‘Generic’ and ‘DoS’; and 10,000 samples each for ‘Reconnaissance,’ ‘Fuzzers,’ and ‘Shellcode.’ These class-specific generation counts increase the representation of underrepresented attacks without imposing equal class sizes.

Attack Class	Samples Generated
Normal	40000
Generic	15000
Exploits	20000
Reconnaissance	10000
Fuzzers	10000
Analysis	35000
Worms	35000
Shellcode	10000
DoS	15000
Backdoor	25000

In this dataset, we have considered all the features for synthetic sample generation and classification. There are 45 features in this dataset, including ‘label,’ ‘attack_class,’ and ‘id,’ which are generally used for model training. We have 3 categorical features like ‘proto’, ‘service’, ‘state’.

label: Class label (0: normal, 1: attack).

attack_cat: Name of specific attack class (*Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shellcode, Worms, and Normal*).

Following are the details of each feature:

Flow & Basic Features

- **id:** Row identifier number.
- **dur:** Total record transaction duration (in seconds).
- **proto:** Transaction protocol (e.g., TCP, UDP, ICMP).
- **service:** App protocol used (e.g., http, ftp, smtp, dns, or - for none).
- **state:** Connection state and its status (e.g., INT, FIN, CON).
- **spkts:** packet count from Source-to-destination.
- **dpkts:** Destination-to-source packet count.
- **sbytes:** Source-to-destination transaction bytes.
- **dbytes:** Destination-to-source transaction bytes.
- **rate:** Packet transmission rate (packets per second).
- **sttl:** Source-to-destination time-to-live value.
- **dttl:** Destination-to-source time-to-live value.
- **sload:** Source-to-destination bit load (bandwidth per second).

- **dload**: bit load from destination to source (bandwidth per second).
- **sloss**: Source packets retransmitted or dropped.
- **dloss**: Destination packets retransmitted or dropped.

Content Features (Payload/Connection Structure)

- **sinpkt**: Source-to-destination inter-packet arrival time (ms).
- **dinpkt**: Destination-to-source inter-packet arrival time (ms).
- **sjit**: Source jitter (packet delay variance) (ms).
- **djit**: Destination jitter (packet delay variance) (ms).
- **swin**: Source TCP window size.
- **stcpb**: TCP source base sequence number.
- **dtepb**: base sequence TCP number for the destination.
- **dwin**: Destination TCP window size.
- **tcprtt**: The SYN-ACK TCP round trip time.
- **synack**: Time from SYN to SYN-ACK packets
- **ackdat**: Time from SYN-ACK to ACK packets.
- **smean**: mean of packet size sent by the source.
- **dmean**: Average packet size sent by the destination.
- **trans_depth**: depth of transaction in HTTP request/response pipelines.
- **response_body_len**: Actual HTTP response body file size (bytes).

Time & General Purpose Features

- **stime**: Record start timestamp.
- **ltime**: Record last transaction timestamp.
- **ct_dst_src_ltm**: Count of connections sharing identical source and destination IPs.
- **is_ftp_login**: Binary flag (1 if FTP session is accessed/logged in, 0 otherwise).
- **is_sm_ips_ports**: Binary flag 1 if source and destination IP/ports match it is 1, else 0.
- **ct_state_ttl**: Connections matching counts for specific state and TTL combinations.
- **ct_flw_http_mthd**: Count of flows using HTTP methods (like GET/POST).
- **ct_dst_sport_ltm**: Connections count in targeting the same destination port from the same source port.
- **ct_ftp_cmd**: FTP commands count utilized in the session.
- **ct_srv_src**: Count of connections sharing the same service and source address.
- **ct_srv_dst**: Count of connections sharing the same service and destination address.
- **ct_dst_ltm**: Connections count targeting the same destination host over a recent timeframe.

- **ct_src_ltm**: Connections count originating from the same source host over a recent timeframe.
- **ct_src_dport_ltm**: Connections count from the same source address targeting the same destination port.